

# GEOGRAPHY

## INTRODUCTION

Geography is about determining the significance of “place” as it relates to the natural environment, the human environment, and interactions within and between them. To investigate geographic issues, students must analyse the influences and interrelationships that give a place its distinctive characteristics and thus its spatial importance. Geographic analysis also requires an investigation of the economic, environmental, social, and political perspectives that relate to an issue. The application of the concepts of geographic thinking, spatial skills, and the use of field studies are central to the geographic inquiry process and the learning of geography.

## Strands

The Grade 11 and 12 geography courses each have five strands. In all of these courses, strand A, Geographic Inquiry and Skill Development, is followed by four content strands, which are organized thematically.

### **Citizenship Education**

The expectations in the Grade 11 and 12 geography courses provide opportunities for students to explore a number of concepts connected to the citizenship education framework (see page 13).

## The Concepts of Geographic Thinking

The four concepts of geographic thinking – spatial significance, patterns and trends, interrelationships, and geographic perspective – underpin thinking and learning in all geography courses in the Canadian and world studies program. At least one concept of geographic thinking is identified as the focus for each overall expectation in the content strands of these courses. The following chart describes each concept and provides sample questions related to it. These questions highlight opportunities for students to apply a specific concept in their studies. (See page 16 for a fuller discussion of the concepts of disciplinary thinking.)

| Spatial Significance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| <p>This concept requires students to determine the importance of a place or region. They explore the connections that exist between the geographical location and physical characteristics of a site and analyse the unique relationships that exist in and between the natural and human environments in a particular place. Students come to understand that the significance of the same place may be different for humans, animals, and plants.</p> <p><b>Related Questions*</b></p> <ul style="list-style-type: none"> <li>– What does a population settlement map tell us about the preferred range of latitude for settlement in Canada? (CGC1P, B2.2)</li> <li>– How does the official plan for your community address urban sprawl? (CGC1D, E2.3)</li> <li>– How might the uniqueness of a destination make it more appealing to some and less appealing to others? (CGG3O, B1.2)</li> <li>– How does the location of a site influence the feasibility of different types of power generation there? (CGR4E, E3.4)</li> <li>– What are some of the challenges that communities in permafrost regions face in providing safe drinking water and disposing of waste? (CGR4M, E2.3)</li> </ul> |
| Patterns and Trends                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <p>This concept requires students to recognize characteristics that are similar and that repeat themselves in a natural or human environment (patterns) and characteristics or traits that exhibit a consistent tendency in a particular setting over a period of time (trends). The characteristics may be spatial, social, economic, physical, or environmental. Students analyse connections between characteristics to determine patterns; they analyse connections between those characteristics over time to determine trends.</p> <p><b>Related Questions</b></p> <ul style="list-style-type: none"> <li>– What trends do you see in the use of alternative energy in Ontario? (CGC1P, A1.5)</li> <li>– What pattern or patterns do you see in the location of First Nations reserves across Canada? (CGC1D, D3.1)</li> <li>– What factors have led to population movement in the region? What specific locations have attracted migrants? Why? (CGD3M, E2.1)</li> <li>– In what ways has technological change amplified the impact of population growth on the natural environment? In what ways has it diminished it? (CGW4U, C2.2)</li> </ul>                                              |

\* These questions are drawn directly from the overview charts that precede the geography courses and from the sample questions that accompany many specific expectations. To highlight the continuity between the geography courses in Grade 11 and 12 and those in Grade 9, and to show possible development in the use of the concepts of geographic thinking over those grades, the chart includes some questions from the Grade 9 geography curriculum as well.

### Interrelationships

This concept requires students to explore connections within and between natural and human environments. The interconnected parts of an environment or environments work together to form a system. Students must understand the relationships that exist within a system and then critically analyse the relationships between systems in order to determine the impact they have on one another.

#### Related Questions

- How does surrounding farmland support a community, and what stresses might the community place on the farmland? (CGC1P, E2.1)
- How might the breaking up of continental ice in Greenland and the Antarctic affect Canada's coastline? (CGC1D, B2.1)
- Why do we need to understand the earth's physical processes in order to predict their potential hazards? (CGF3M, Overview)
- How can international trade policies and corporate actions influence disparities within or between ecumenes? (CGU4M, D1.2)

### Geographic Perspective

This concept requires students to consider the environmental, economic, political, and/or social implications of the issues, events, developments, and/or phenomena that they are analysing. In order to solve problems, make decisions or judgements, or formulate plans of action effectively, students need to develop their ability to examine issues from multiple perspectives.

#### Related Questions

- In what ways can cultural diversity enrich the life of a community? (CGC1P, D1.2)
- Does the financial benefit of extracting natural resources justify related social and/or environmental impacts? (CGC1D, A1.6)
- How might spatial technologies help governments assess the economic, environmental, infrastructural, and social risks of a potential disaster? Why is it important for governments to know and understand these risks? (CGT3O, D1.2)
- Why might some people believe that there is an ethical imperative to share the world's resources equitably? (CGW4C, Overview)
- Why might military and/or government satellite information be shared with or withheld from certain countries or military alliances? What are the implications of doing so? (CGO4M, D2.1)

## The Geographic Inquiry Process

In each of the geography courses in the Canadian and world studies curriculum, strand A focuses explicitly on the geographic inquiry process, guiding students in their investigations of issues, events, developments, and/or various geographic phenomena. This process is *not* intended to be applied in a linear manner: students will use the applicable components of the process in the order most appropriate for them and for the task at hand. Although strand A covers all of the components of the inquiry process, it is important to note that students apply skills associated with the inquiry process throughout the content strands in each course. (See page 32 for a fuller discussion of the inquiry process in the Canadian and world studies program.)

The following chart identifies ways in which students may approach each of the components of the geographic inquiry process.

| Formulate Questions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| <p>Students formulate questions:</p> <ul style="list-style-type: none"> <li>– to explore various events, developments, issues, and/or phenomena that are related to the overall expectations in order to identify the focus of their inquiry</li> <li>– to help them determine which key concept or concepts of geographic thinking are relevant to their inquiry</li> <li>– that reflect the selected concept(s) of geographic thinking</li> <li>– to develop criteria that they will use in evaluating data, evidence, and/or information; in making judgements, decisions, or predictions; in reaching conclusions; and/or in solving problems</li> </ul>                                                                                                                                                                          |
| Gather and Organize                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <p>Students:</p> <ul style="list-style-type: none"> <li>– collect relevant qualitative and quantitative data, evidence, and information from field studies<sup>a</sup> and a variety of primary and secondary sources,<sup>b</sup> including visuals<sup>c</sup> and community resources<sup>d</sup></li> <li>– determine if their sources are credible, accurate, and reliable</li> <li>– identify the purpose and intent of each source</li> <li>– identify the points of view in the sources they have gathered</li> <li>– use a variety of methods to organize the data, evidence, and/or information they have gathered</li> <li>– record the sources of the data, evidence, and information they are using</li> <li>– decide whether they have collected enough data, evidence, and/or information for their inquiry</li> </ul> |

a. Field studies may include, but are not limited to, studies in local neighbourhoods, school grounds, and various sites that allow students to explore different land uses (e.g., recreational, commercial, industrial, and transportation uses).

b. Primary sources may include, but are not limited to, census data, land claims, letters, photographs, speeches, and works of art. Secondary sources may include, but are not limited to, documentaries and other films, news articles, reference books, and most websites.

c. Visuals may include, but are not limited to, satellite images, maps, globes, models, graphs, and diagrams.

d. Community resources may include, but are not limited to, local conservation areas, resources from community groups and associations, government resources, and local plans.

| Interpret and Analyse                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Students:</p> <ul style="list-style-type: none"> <li>– analyse data, evidence, and information, applying the relevant concepts of geographic thinking (see preceding chart)</li> <li>– use different types of graphic organizers to help them interpret and/or analyse their data, evidence, and information</li> <li>– identify the key points or ideas in each source</li> <li>– analyse graphs, charts, diagrams, and maps</li> <li>– construct graphs, charts, diagrams, and maps to help them analyse the issue, event, development, or phenomenon they are investigating</li> <li>– analyse their sources to determine the importance of an issue, event, development, or phenomenon for individuals or groups</li> <li>– identify biases in individual sources</li> <li>– determine if all points of view are represented in the source materials as a whole, and which, if any, are missing</li> </ul> |
| Evaluate and Draw Conclusions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <p>Students:</p> <ul style="list-style-type: none"> <li>– synthesize data, evidence, and information, and make informed, critical judgements based on that data, evidence, and information</li> <li>– determine the short- and long-term impact of an event, development, issue, or phenomenon on people and/or places</li> <li>– reach conclusions about the subject of their inquiry, and support them with their data, evidence, and information</li> <li>– make predictions based on their data, evidence, and information</li> <li>– determine the ethical implications of an issue or action</li> <li>– determine the action required, where appropriate</li> </ul>                                                                                                                                                                                                                                         |
| Communicate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <p>Students:</p> <ul style="list-style-type: none"> <li>– use appropriate forms (e.g., oral, visual, written, multimedia) for different audiences and purposes</li> <li>– communicate their arguments, conclusions, predictions, and plans of action clearly and logically</li> <li>– use geographical terminology and concepts correctly and effectively</li> <li>– cite sources, using appropriate forms of documentation</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

# Introduction to Spatial Technologies, Grade 11

Open

CGT30

This course enables students to develop practical skills associated with spatial technologies and to investigate related career opportunities. Students will develop their ability to use geographic information systems (GIS), global positioning systems (GPS), and remote sensing technologies. They will apply the concepts of geographic thinking and the geographic inquiry process when conducting fieldwork, collecting and organizing data, and analysing spatial images such as maps and aerial photographs. Throughout the course, students' local context is emphasized.

**Prerequisite:** Issues in Canadian Geography, Grade 9, Academic or Applied

## OVERVIEW

The course has five strands. Instruction and learning related to the expectations in strand A are to be interwoven with instruction and learning related to expectations from the other four strands. Strand A must not be seen as independent of the other strands. Student achievement of the expectations in strand A is to be assessed and evaluated *throughout* the course.

### Strand A

| A. Geographic Inquiry and Skill Development                                                                                                                                                                                          |
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| Overall Expectations                                                                                                                                                                                                                 |
| <b>A1. Geographic Inquiry:</b> use the geographic inquiry process and the concepts of geographic thinking when conducting investigations using spatial technologies                                                                  |
| <b>A2. Developing Transferable Skills:</b> apply in everyday contexts skills, including spatial skills, developed through geographical investigation, and identify some careers in which a background in geography might be an asset |

(continued)

**Overview** *(continued)*

Throughout this course, when planning instruction, teachers should weave the expectations from strand A in with the expectations from strands B–E.

**Strands B–E**

| Overall Expectations and Related Concepts of Geographic Thinking                                                                                                                                                                                                                                                                            | Big Ideas*                                                                                                       | Framing Questions*                                                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>B. Spatial Geography: Concepts and Processes</b>                                                                                                                                                                                                                                                                                         |                                                                                                                  |                                                                                                                                                          |
| <b>B1. Applying Skills in a Local Context:</b> use a variety of spatial technologies to collect and analyse information and field data for the purpose of assessing local geography and geographic issues ( <b>FOCUS ON:</b> <i>Spatial Significance; Geographic Perspective</i> )                                                          | Spatial technologies are essential tools for making decisions or informed judgements on local geographic issues. | In what ways can spatial technologies be used when investigating a local development project?                                                            |
| <b>B2. Fundamentals of Field Data Collection:</b> demonstrate an understanding of the fundamental concepts and processes associated with the collection and analysis of field data using a variety of spatial technologies ( <b>FOCUS ON:</b> <i>Spatial Significance; Patterns and Trends</i> )                                            | Spatial technologies allow people to collect and analyse large amounts of various types of geographical data.    | Which spatial technologies would help you conduct a local field study?                                                                                   |
| <b>B3. Fundamentals of Cartography:</b> describe and use basic mapping conventions, various types of maps, and fundamental concepts of cartography, and apply that knowledge to analyse bias in the construction and use of maps ( <b>FOCUS ON:</b> <i>Interrelationships; Geographic Perspective</i> )                                     | Mapping conventions allow people to decode the information contained in maps.                                    | Is there a language of mapping? Why do some maps not follow the conventions?<br><br>How can a map or graph be biased (intentionally or unintentionally)? |
| <b>C. Using Spatial Technologies to Support Sustainability and Stewardship</b>                                                                                                                                                                                                                                                              |                                                                                                                  |                                                                                                                                                          |
| <b>C1. Creating and Supporting Sustainable Spaces:</b> use a variety of spatial technologies to help them assess human activities and plan and promote the sustainable use of the natural environment, including natural resources, in their local community or area ( <b>FOCUS ON:</b> <i>Interrelationships; Geographic Perspective</i> ) | Spatial technologies are key tools in planning for sustainability.                                               | Why is it important for local companies and governments to use spatial technologies when making decisions about the development of the community?        |
| <b>C2. Analysing Environmental Issues:</b> use a variety of spatial technologies to analyse the impact of human activity on the environment in their local community or area and beyond, and identify possible solutions ( <b>FOCUS ON:</b> <i>Spatial Significance; Patterns and Trends</i> )                                              | Spatial technologies are key tools in the analysis and resolution of local environmental issues.                 | How can spatial technologies be used to help promote environmental sustainability in a community?                                                        |

\* See page 17 for a discussion of the purpose of big ideas and framing questions.

| Overall Expectations and Related Concepts of Geographic Thinking                                                                                                                                                                                                                                                                             | Big Ideas*                                                                                                              | Framing Questions*                                                                                                                                                                                                                                     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>D. Spatial Technologies, Society, and Interdependence</b>                                                                                                                                                                                                                                                                                 |                                                                                                                         |                                                                                                                                                                                                                                                        |
| <b>D1. Spatial Technologies and Society:</b> describe and assess the ways in which different groups in society use spatial technologies for planning and decision making ( <b>FOCUS ON:</b> <i>Interrelationships; Geographic Perspective</i> )                                                                                              | Different groups use spatial technologies for different reasons and under different circumstances.                      | Under what circumstances might the use of spatial technologies be a source of disagreements or conflict?                                                                                                                                               |
| <b>D2. Connecting Global and Local Issues:</b> use a variety of spatial technologies to analyse global issues related to geography and their impacts on the local community or area ( <b>FOCUS ON:</b> <i>Patterns and Trends; Interrelationships</i> )                                                                                      | Spatial technologies allow people to analyse global patterns and determine the connection to their local area.          | How might spatial technologies be used by different groups to support their own arguments, plans, or goals around the same issue?<br><br>How can spatial technologies be used to help understand the impact of global weather systems on a local area? |
| <b>E. Spatial Technologies and Change</b>                                                                                                                                                                                                                                                                                                    |                                                                                                                         |                                                                                                                                                                                                                                                        |
| <b>E1. Monitoring Local Change:</b> investigate the use of spatial technologies to monitor trends and predict change at the local level ( <b>FOCUS ON:</b> <i>Patterns and Trends; Geographic Perspective</i> )                                                                                                                              | Spatial technologies allow people to study how and why change occurs and to use this understanding to make predictions. | What can you learn from conducting a longitudinal study of natural and human change in your area? Who might benefit from this information?                                                                                                             |
| <b>E2. Development of Spatial Technologies:</b> describe developments in spatial technologies over time, including Canadian contributions to the field, and explain some of the effects of these developments on job opportunities and on aspects of the broader society ( <b>FOCUS ON:</b> <i>Patterns and Trends; Interrelationships</i> ) | Spatial technology is being used in almost every type of workplace.                                                     | How might spatial technologies aid in understanding the impact of the spread of a disease on a local area?<br><br>How have the purpose and design of spatial technologies changed over time? Why have these changes occurred?                          |

# A. GEOGRAPHIC INQUIRY AND SKILL DEVELOPMENT

## OVERALL EXPECTATIONS

Throughout this course, students will:

- A1. Geographic Inquiry:** use the geographic inquiry process and the concepts of geographic thinking when conducting investigations using spatial technologies;
- A2. Developing Transferable Skills:** apply in everyday contexts skills, including spatial skills, developed through geographical investigation, and identify some careers in which a background in geography might be an asset.

## SPECIFIC EXPECTATIONS

### A1. Geographic Inquiry

Throughout this course, students will:

**A1.1** formulate different types of questions to guide investigations using spatial technologies (e.g., factual questions: *How is vegetation health identified on an infrared photograph?*; comparative questions: *Which map scale is most effective for identifying a transportation route?*; causal questions: *What factors might cause the loss or degradation of a GPS signal?*)

**A1.2** select and organize relevant data and information on geographic issues from a variety of primary and secondary sources (e.g., primary: *raw data from fieldwork, both quantitative and qualitative; photographs; satellite images*; secondary: *published statistics, newspapers, books, atlases, geographic magazines, websites, graphs, charts, digital and print maps*), ensuring that their sources represent a diverse range of perspectives

**Sample questions:** “How might you use statistics on incidence of disease to analyse the pattern of spread of a chosen disease in different parts of the world? Where might you find this data and information?” “What type of data and information do you need to collect in order to assess the impact of a storm that caused flooding in a city?” “What types of maps and graphs will help you analyse immigration patterns?”

**A1.3** assess the credibility of sources and information relevant to their investigations (e.g., *by considering how the data are constructed to support the author’s point of view, the possible bias of the author, the expertise of the author, the accuracy of the text and supporting data, the intended audience, the purpose of the messaging, the context in which the information was presented*)

**Sample questions:** “What is the source of this map or photographic image? What biases might this source have and how might they affect the map or image? Have you consulted other maps or images of the same place from a different source or a slightly different time period? How do they compare?”

**A1.4** interpret and analyse data and information relevant to their investigations, using various tools, strategies, and approaches appropriate for geographic inquiry (e.g., *interpret diagrams illustrating the flow of different products within an economic system; analyse graphs and charts of quality of life indicators for selected countries to determine trends and correlations; use decision-making templates to analyse points of view on a local issue; use graphic organizers to outline various perspectives on natural resource use in their region*)

**Sample questions:** “What type of graphic organizer would you use to help analyse the impact of offshore production on both the manufacturing country and the consumer country?” “What data layers and types of information might you include when using GIS to analyse global production of greenhouse gases?”

**A1.5** use the concepts of geographic thinking (i.e., spatial significance, patterns and trends, interrelationships, geographic perspective) when analysing and evaluating data and information, formulating conclusions, and making judgements about issues they are investigating through the use of spatial technologies (e.g., *use the concept of spatial significance to analyse a map layer showing the distribution of desertification or poverty; use the concept of patterns and trends to analyse tables showing short- and long-term population trends and phenomena such as urbanization; use the concept of interrelationships to explain how a variety of natural and human factors depicted in map layers combine to contribute to global inequality; use the concept of geographic perspective to analyse the social, political, economic, and environmental impacts of the use of GPS technology*)

**Sample questions:** “How can an understanding of spatial significance help you when determining the type of physical environment in which you would find wetlands?” “How might an understanding of patterns and trends help you analyse a satellite image to determine the impact of the melting of continental ice?” “How might an understanding of interrelationships guide your analysis of a story map showing the connections among the following: a product you buy, the company that sells it, the resources used to make and dispose of it, and labour conditions in the factory that manufactures it?” “How can geographic perspective help you analyse the impact of zoning on community safety by layering maps that depict types of criminal incidents with those that show various types of land use, such as residential, commercial, and recreational uses?”

**A1.6** evaluate and synthesize their findings to formulate conclusions and/or make informed judgements or predictions about the issues they are investigating

**Sample questions:** “What did you find out about the relationship between trends in river flooding and the amount of damage done to nearby buildings? What conclusions can you draw about the impact of this relationship on the area or region? How might this relationship affect city planning in the future, such as the location of new subdivisions in relation to existing or planned floodways or floodplains?”

**A1.7** communicate their ideas, arguments, and conclusions using various formats and styles, as appropriate for the audience and purpose (e.g., *a debate for classmates on which map projection is best to use for a specific country; a video for a*

*Grade 9 class showing how data on a community issue can be gathered and then presented in map form; a webinar or podcast for the general public on how their local area connects to world systems on a daily basis; an annotated map for a local community group to illustrate the diversity of cultures within their community*)

**Sample questions:** “What kind of information does your audience need?” “What geographic scale should you use in order to show the level of detail that you need to support your argument?” “What format and approach would be most effective in conveying your information to this particular audience?”

**A1.8** use accepted forms of documentation (e.g., footnotes, author/date citations, reference lists, bibliographies, annotated bibliographies, credits) to reference different types of sources (e.g., websites, blogs, books, articles, films, data)

**A1.9** use appropriate terminology when communicating the results of their investigations (e.g., *vocabulary specific to their inquiry; terminology related to geography and to the concepts of geographic thinking*)

## A2. Developing Transferable Skills

Throughout this course, students will:

**A2.1** describe ways in which geographic investigation can help them develop skills, including spatial skills and the essential skills in the Ontario Skills Passport (e.g., *reading graphic texts, writing, graphing, computer use, use of spatial technologies, oral communication, numeracy, decision making, planning, management, finding information, problem solving*), that can be transferred to postsecondary opportunities, the world of work, and everyday life

**A2.2** apply in everyday contexts skills and work habits developed through geographic investigation (e.g., *ask questions to deepen their understanding of an issue; use quantitative data to support an idea; use spatial skills to aid their understanding of world events; apply work habits such as collaboration, listening to and considering multiple perspectives when discussing an issue as part of a group*)

**A2.3** apply the concepts of geographic thinking when analysing current events involving geographic issues (e.g., *use the concept of spatial significance to analyse possible reasons for the spread of a disease; use the concept of patterns*

*and trends to analyse potential impacts of El Niño and La Niña on North and South America; use the concept of interrelationships to analyse the connection between high poverty levels and civil unrest in a country; use the concept of geographic perspective to analyse the potential consequences of a change in political leadership in a country) in order to enhance their understanding of these issues and their role as informed citizens*

**A2.4** identify some careers in which a geography background might be an asset (*e.g., civil engineer, forestry analyst, oceanographer, transportation planner, emergency planning specialist, military personnel, border services officer, real estate agent, land surveyor, marketing analyst, town planner, GIS technician*)

# B. SPATIAL GEOGRAPHY: CONCEPTS AND PROCESSES

## OVERALL EXPECTATIONS

By the end of this course, students will:

- B1. Applying Skills in a Local Context:** use a variety of spatial technologies to collect and analyse information and field data for the purpose of assessing local geography and geographic issues (**FOCUS ON:** *Spatial Significance; Geographic Perspective*)
- B2. Fundamentals of Field Data Collection:** demonstrate an understanding of the fundamental concepts and processes associated with the collection and analysis of field data using a variety of spatial technologies (**FOCUS ON:** *Spatial Significance; Patterns and Trends*)
- B3. Fundamentals of Cartography:** describe and use basic mapping conventions, various types of maps, and fundamental concepts of cartography, and apply that knowledge to analyse bias in the construction and use of maps (**FOCUS ON:** *Interrelationships; Geographic Perspective*)

## SPECIFIC EXPECTATIONS

### B1. Applying Skills in a Local Context

**FOCUS ON:** *Spatial Significance; Geographic Perspective*

By the end of this course, students will:

- B1.1** use various types of maps (*e.g., topographic, choropleth, thematic, relief*) of different places with similar features to analyse the unique characteristics of such features in their local community or area

**Using spatial skills:** Students can explore online GIS base maps illustrating various features of their local community or area and describe situations where one would use such a map.

- B1.2** design and conduct a structured study, using the geographic inquiry process, to address a need in their local community or area (*e.g., investigate possible locations for a new outdoor sports complex geared towards young adults, investigate the impact of existing industrial development near a protected habitat and river system*), and use a variety of spatial technologies to collect and analyse the data (*e.g., use GIS to layer themes and data to analyse the locations of chain restaurants in their community, use remote sensing data to identify possible locations of mineral sources or track the spread of an invasive insect*)

**Sample questions:** “What is the best location in our community for a specific business or service, such as a coffee shop, a pizza delivery service, a bank branch, or an ATM?” “Where might there be a need for a sports and recreation facility such as an arena, public beach, marina, baseball diamond, miniature golf course, skateboard park, zip-line or high-ropes course, splash pad, or paintball complex? Why?” “What is the best location for a search-and-rescue or emergency-services headquarters in our community?” “If a business starts a bike- or car-share program in our community, where should the bike or car pickup locations be?”

- B1.3** use a variety of spatial technologies to investigate the impact of a real or potential local development project on the surrounding physical geography (*e.g., examine aerial photographs to identify existing deforested areas; use remote sensing to identify the glacial features of a region that need protection; use GIS to study river systems and run-off patterns near a proposed industrial site or to identify hazard-prone areas prior to construction*)

**Sample questions:** “What geographic features might be considered for preservation from development?” “What are the most hazard-prone areas in the proposed subdivision?” “How might the use of remote sensing help in assessing the

proposed location of a new ski resort, including possible environmental issues at the proposed site?”

**Using spatial skills:** Students can use the spatial analysis tools in GIS to create a series of raster datasets that illustrate the characteristics of a specific physical site.

## B2. Fundamentals of Field Data Collection

**FOCUS ON:** *Spatial Significance; Patterns and Trends*

By the end of this course, students will:

**B2.1** identify and describe various types of data used to create images and construct maps with the aid of spatial technologies (*e.g., primary versus secondary data; analog versus digital data; sets of layers or themes; spatial versus non-spatial data; attribute data; raster versus vector data; lines and polygons in GIS; qualitative versus quantitative data*)

**B2.2** use GPS to collect and analyse location and elevation data on selected geographic features in their local community or area

**Sample question:** “How can GPS be used to determine: the route of a trail; property boundaries; the location of historical landmarks, hydrological features, or a geocache?”

**B2.3** use a variety of spatial technologies to gather data through a local field study or survey, code and tabulate the data (*e.g., by performing tasks such as buffering data, querying, geocoding, georeferencing, creating new spreadsheet fields, building attribute tables*), and interpret the results

**Sample questions:** “What patterns do you see emerging from the data or survey results? What new questions do you have based on these patterns?”

**B2.4** use a database they have built (*e.g., using qualitative and quantitative attribute data; combining their own data with existing data from other sources*) to produce a variety of thematic maps, charts, and graphs illustrating aspects of local geography

**Using spatial skills:** Students can construct a map layout using GIS.

## B3. Fundamentals of Cartography

**FOCUS ON:** *Interrelationships; Geographic Perspective*

By the end of this course, students will:

**B3.1** identify common mapping conventions (*e.g., title, legend, scale, projection and datum, orientation, author, date*) and use them appropriately when analysing and constructing a variety of maps and charts

**B3.2** differentiate between large scale and small scale maps and their functions, and use scale to calculate distance and area and to assess the degree of exaggeration and/or accuracy of various maps

**Sample questions:** “Why is it important to understand the concept of scale when reading two different maps of an area? Why is it important to identify the purpose behind the construction of each map?” “Why might two different stakeholders choose to produce maps of a proposed highway development project using two different scales? Why might a construction company or a developer want a less detailed (smaller-scale) map of the area? Why might a wildlife protection group want a more detailed (larger-scale) map that shows the specific habitats and natural features of the area?”

**B3.3** explain the concept of direction (*e.g., with reference to true north, magnetic north, bearings, grid directions, “look direction”, range, total field of view, flight path*) and various aspects of its use in cartography, GIS, remote sensing, and GPS

**Sample questions:** “Why is it important to understand the difference between true north and magnetic north? How might mistaking them for each other be a problem when using a topographic map?”

**B3.4** identify and describe some common map projections (*e.g., Mercator [cylindrical], Peters [equal-area cylindrical], Lambert [conical], polar [azimuthal]*), and analyse the limitations and biases of each

**Sample questions:** “How are various continents and countries presented in different map projections?” “How does the choice of map projections influence the message being conveyed? How does this choice reveal media bias?”

**B3.5** explain the concept of location with the aid of a variety of methods and tools (*e.g., geographic coordinates [latitude and longitude], grid coordinates [the Universal Transverse Mercator (UTM) system, geodetic datum], geocoding [using street address, postal code], georeferencing, geocaching, triangulation, trilateration*)

*Sample question:* “How might the way you choose to determine or explain location depend on the type of map or tool you are using?”

**B3.6** explain the fundamental concepts and processes of a geographic information system (GIS), global positioning system (GPS), and remote sensing, using appropriate terminology (*e.g., for GIS: attribute table, layering, querying, buffering; for GPS: satellite and receiver, radio waves, elevation, triangulation, trilateration; for remote sensing: electromagnetic spectrum, targets, active and passive sensors, satellites*)

# C. USING SPATIAL TECHNOLOGIES TO SUPPORT SUSTAINABILITY AND STEWARDSHIP

## OVERALL EXPECTATIONS

By the end of this course, students will:

- C1. Creating and Supporting Sustainable Spaces:** use a variety of spatial technologies to help them assess human activities and plan and promote the sustainable use of the natural environment, including natural resources, in their local community or area (**FOCUS ON:** *Interrelationships; Geographic Perspective*)
- C2. Analysing Environmental Issues:** use a variety of spatial technologies to analyse the impact of human activity on the environment in their local community or area and beyond, and identify possible solutions (**FOCUS ON:** *Spatial Significance; Patterns and Trends*)

## SPECIFIC EXPECTATIONS

### C1. Creating and Supporting Sustainable Spaces

**FOCUS ON:** *Interrelationships; Geographic Perspective*

By the end of this course, students will:

- C1.1** plan a hypothetical local development, using spatial technologies as appropriate, in a way that minimizes impact on the environment

**Sample questions:** “What characteristics of the local environment are important to consider when analysing the potential impacts of a development such as a recreational facility, a commercial outlet, an institutional facility, or an industrial complex?” “Why is an understanding of physical geography an important basis for assessing the impacts of a local development?” “How might the use of remote sensing help assess the possible location for a new fishing lodge? How could remote sensing help identify possible environmental issues at the proposed site?”

**Using spatial skills:** Students can use GIS to create a hypothetical development in a given area. They can add shaded polygons to symbolize various types of land use. Imagery can be added to illustrate specific features of the local physical environment or of the potential development. Buffers can be added, using the measuring tool, to help determine the area of the impact that the development will have on the physical environment.

- C1.2** describe the role of spatial technologies in assessing the sustainability of a variety of outdoor recreational activities and venues (e.g., activities such as hiking, fishing, snowmobiling, canoeing, boating, or kayaking; venues such as a ski resort, a cottage development, a golf course, or a provincial park)

**Sample questions:** “How might the use of remote sensing and GIS help assess whether the route of an existing snowmobile trail should be changed? How might the use of remote cameras help you determine whether the trail is used by wildlife?”

- C1.3** use spatial technologies to construct a map to support the sustainable use of a natural feature in their local community or area (e.g., a map that shows the least damaging route for a suggested nature trail; a map that illustrates a river course and the proposed location for a housing development; a story map showing a shoreline and identifying the possible locations for public access that will not damage or further erode the shoreline; an annotated map that explains the environmental necessity of a moraine and some possible sustainable uses for the moraine)

**Sample questions:** “What characteristics of the place you have chosen will you include on your map? Why?”

**Using spatial skills:** Students can construct a “story map” to promote the sustainable use of

a natural feature or area to a general audience. When different types of media, such as photos and videos, are linked with locations, people viewing the story map can get a more vivid impression of those locations.

**C1.4** describe the role of spatial technologies in facilitating the sustainable use of natural resources (*e.g., natural resource extraction; use of renewable and non-renewable energy sources; business and industrial practices involving forests, minerals, fisheries, wetlands, and/or lakes*), with a focus on their local community or area

**Sample questions:** “How can spatial technologies be used to determine the best location for a furniture manufacturer that wants to reduce its energy consumption related to transportation?” “How might spatial technologies help us choose the best location in our community for homes that will use geothermal heating and cooling?”

## C2. Analysing Environmental Issues

**FOCUS ON:** *Spatial Significance; Patterns and Trends*

By the end of this course, students will:

**C2.1** analyse issues related to human use of the local environment, using a variety of spatial technologies (*e.g., by gathering data through fieldwork or surveys, analysing the data to assess the impact of human activities, and drawing conclusions or making recommendations based on the analysis*)

**Sample question:** “How would you gather and analyse data to assess the impact on the local environment of activities such as tree planting, landscaping an outdoor classroom, the creation and use of walking trails, garbage disposal practices, or the creation of a skate park?”

**Using spatial skills:** Students can collect field data using a device such as a GPS unit, a smartphone, or a tablet and construct a map to illustrate the local area and determine areas for improvement. Field data may include photographs, videos, locational information (*e.g., latitude and longitude, addresses*), measurements, and/or text.

**C2.2** analyse remotely sensed images to obtain a synoptic view of selected human activities (*e.g., agricultural land use, urbanization, deforestation*) that affect the natural environment (*e.g., through species extinction, insect infestation*)

**Sample question:** “What techniques can you use to interpret remote sensing images to analyse levels of deforestation?”

**C2.3** use GPS or GIS to analyse the impacts of development on selected local water resources (*e.g., erosion, water withdrawal, changes in water quality; effects of sewage treatment, beach maintenance*), and propose solutions based on their analysis

**Sample question:** “In what ways can a development or water diversion affect water systems beyond the original site?”

**Using spatial skills:** Students can use GIS to create a display of their local area and identify water resources. Using the measuring tool, students can add buffers to determine areas where development may be affecting the water resources they have identified.

# D. SPATIAL TECHNOLOGIES, SOCIETY, AND INTERDEPENDENCE

## OVERALL EXPECTATIONS

By the end of this course, students will:

- D1. Spatial Technologies and Society:** describe and assess the ways in which different groups in society use spatial technologies for planning and decision making (**FOCUS ON:** *Interrelationships; Geographic Perspective*)
- D2. Connecting Global and Local Issues:** use a variety of spatial technologies to analyse global issues related to geography and their impacts on the local community or area (**FOCUS ON:** *Patterns and Trends; Interrelationships*)

## SPECIFIC EXPECTATIONS

### D1. Spatial Technologies and Society

**FOCUS ON:** *Interrelationships; Geographic Perspective*

By the end of this course, students will:

- D1.1** describe and assess some of the ways in which spatial technologies are used to support community affairs and public services (*e.g., GPS is used to monitor criminal activity; satellites are used to gather data for monitoring of common urban issues*), and identify potential issues related to these uses

**Sample questions:** “How can GIS help with mapping criminal activity in a community? What are the implications of GPS and GIS use by police?” “Why is it important for fire departments and paramedics to use some spatial technologies? Which ones would be most helpful for emergency services, and why?” “What are some of the negative aspects of the use of spatial technologies in a community context?” “Do all communities have access to these technologies? Why or why not? How does inequality of access affect the communities that do and do not have access?”

- D1.2** describe and assess the role of spatial technologies in risk assessment and disaster planning

**Sample questions:** “How can spatial technologies be used effectively by local, regional, and national governments when planning for disaster ‘what-if’ scenarios or performing risk identification and assessment?” “How do spatial technologies

help people understand and assess the risks of a tsunami in a given region?” “How might spatial technologies help governments assess the economic, environmental, infrastructural, and social risks of a potential disaster? Why is it important for governments to know and understand these risks?”

- D1.3** describe and assess the role of spatial technologies in informing human responses to natural and human-made disasters (*e.g., natural disasters: floods, tornadoes, earthquakes, snow and ice storms, heat waves; human-made disasters: chemical spills, transport accidents*)

**Sample questions:** “How can spatial technologies be used effectively in disaster response and evacuations? What are the limitations of spatial technologies in relation to disaster relief, recovery, and relocation of affected populations?” “How can the use of GIS and GPS help improve emergency response times?”

- D1.4** describe and assess some of the ways in which various levels and areas of government use spatial technologies in public-sector decision-making processes (*e.g., land-use surveys and assessments are used in planning for new developments or responding to a request for a change in land-use designation; remote sensing and GIS are used to decide on a route for new water and sewer pipes*)

**Sample questions:** “Why is it important for school boards to use GPS when planning routes for school buses?” “How might spatial technologies

help a municipal government identify environmental or social concerns that could affect the decision to allow a new large-scale industrial complex within the city boundaries?" "How can GIS and GPS help with controlling traffic patterns and relieving traffic congestion?"

- D1.5** describe and assess some of the ways in which businesses use spatial technologies in private-sector decision-making processes (*e.g., to determine flight paths and flight times, to choose locations for new retail complexes or manufacturing centres, to reduce transportation costs*)

**Sample questions:** "How might an insurance company use spatial technologies when conducting insurance assessments?" "How might a real-estate development company use spatial technologies when planning a new mixed-use development?"

- D1.6** describe and assess some of the ways in which various First Nations, Métis, and Inuit communities use spatial technologies to inform decisions around land use and make plans for future development (*e.g., to investigate land claims, to conduct resource inventories, to choose the routes of ice roads or snowmobile trails, to conduct independent reviews of proposals for resource extraction*)

**Sample questions:** "How might the use of spatial technologies help First Nations communities in research and planning concerning the proposed location of a new mine? What types of data and information would they need to collect? Which spatial technology or technologies would help them collect the data and information they need?"

## D2. Connecting Global and Local Issues

### FOCUS ON: *Patterns and Trends; Interrelationships*

By the end of this course, students will:

- D2.1** interpret global maps, remote sensing data (*e.g., from the NASA Earth Observatory website*), and satellite images to analyse relationships between some major physical features of the world, areas of human settlement and activity, and variations in selected climatic variables (*e.g., relationships between mountains and rainfall; human settlement and activity and aerosol size; forest fires, crop burning, and carbon monoxide concentrations; net radiation and net primary production; precipitation and snow cover; water vapour and vegetation; sea surface temperature anomalies and land surface temperature anomalies*), and describe how these relationships affect their local community or area

**Sample questions:** "What do these satellite images reveal about our local ecoregion in relation to the global ecosystem?" "How do satellite images help you to determine how weather systems influence local weather patterns?"

**Using spatial skills:** Students can use GIS to create points, lines, and polygons to illustrate the locations of major physical features around the world.

- D2.2** apply a variety of spatial technologies to identify patterns and trends related to selected global issues, and explain how these trends might affect their local community or area

**Sample question:** "How might climate change affect our local weather, local smog levels, or the spread of non-indigenous species or disease into our local area?"

# E. SPATIAL TECHNOLOGIES AND CHANGE

## OVERALL EXPECTATIONS

By the end of this course, students will:

- E1. Monitoring Local Change:** investigate the use of spatial technologies to monitor trends and predict change at the local level (**FOCUS ON:** *Patterns and Trends; Geographic Perspective*)
- E2. Development of Spatial Technologies:** describe developments in spatial technologies over time, including Canadian contributions to the field, and explain some of the effects of these developments on job opportunities and on aspects of the broader society (**FOCUS ON:** *Patterns and Trends; Interrelationships*)

## SPECIFIC EXPECTATIONS

### E1. Monitoring Local Change

**FOCUS ON:** *Patterns and Trends; Geographic Perspective*

By the end of this course, students will:

- E1.1** identify physical or cultural changes in their local area and/or community over time by analysing a temporal sequence of maps, aerial photographs, or satellite imagery

**Sample questions:** “What differences do you observe in the shape of the local shoreline? How have water levels changed over time?” “What is the impact of urban sprawl on productive soil and agricultural land in our local area?” “How have new transportation corridors changed the patterns of urban land use?” “How have incidents of crime or criminal activities changed in our community over time?” “How has the ethnic make-up of specific neighbourhoods changed over time?” “What is the impact on our community of noise pollution after the construction of an airport or major highway?” “What would be the long-term effects of building an industrial plant in our community? How do you know?”

**Using spatial skills:** Students can construct an animated online map that includes one or more temporal layers, and “play” the map to see how the information changes over time. Students can also use GIS to construct maps comparing two layers by using the swipe or spyglass tool.

- E1.2** assess the role of spatial technologies in monitoring changing features or phenomena and predicting future processes or events at the local level (e.g., *bedbug or termite infestations, amount of street racing, incidence of theft*)

**Sample questions:** “How might spatial technologies be used in marketing analysis and to predict future needs in our community?” “How might spatial technologies help in monitoring the spread of disease over time and informing disease control in our community?” “How might spatial technologies help to assess and predict recreational needs in a new development in our community?”

### E2. Development of Spatial Technologies

**FOCUS ON:** *Patterns and Trends; Interrelationships*

By the end of this course, students will:

- E2.1** describe the development of satellite technology, and explain some of the ways it has changed daily life and how business is conducted in various economic sectors (e.g., *effects on business communications, community disaster planning*)

**Sample questions:** “In what ways have the purpose and design of satellites changed over time?” “What are the limitations and advantages of various types of navigational satellites that

have been used over time?” “How have improvements in the collection of weather data and satellite images helped communities better prepare for large storms?”

**E2.2** describe recent advances in the field of spatial technology (*e.g., the development of Web GIS, which enables more people to use, share, and access information*) and identify some careers that may be created as a result, both globally and locally (*e.g., careers in health care and emergency relief*)

**E2.3** identify key Canadian contributions to the development of spatial technologies over time (*e.g., RADARSAT 1 and 2, Canadian Space Agency [CSA] contributions, Canadian satellites, Canadian contributions to international collaborations, the development of the Prince Albert Satellite Station, Canadian astronauts, the Brewer Ozone Spectrophotometer, various contributions by Natural Resources Canada [NRCan] and Land Information Ontario [LIO]*)